

ForecastHub: Predicting Disease Outbreaks

ARTPARK's ForecastHub is transforming disease forecasting by using existing climate and case data to build AI-based models that predict outbreaks four weeks in advance.

Disease outbreaks such as dengue, malaria and other vector-borne infections continue to challenge public health systems across India. While many outbreaks follow predictable patterns driven by climate, rainfall and environmental conditions, responses often begin only after case numbers start rising.

At ARTPARK (AI and Robotics Technology Park) at the Indian Institute of Science (IISc), Bengaluru, the focus has shifted on enabling the public health system from reacting to outbreaks to predicting them. The organisation has developed an open source ForecastHub, which uses existing climate and case data to build AI-based outbreak prediction models as reusable digital infrastructure, enabling governments, researchers and public health agencies to prepare for outbreaks before they escalate.

From research models to public infrastructure

Most disease forecasting models remain confined to research papers with limited real-world deployment. ARTPARK's objective was to bridge this gap by building a modular platform where forecasting models, datasets and deployment tools can be shared and reused across multiple regions for various diseases.

The initiative began with dengue forecasting in Karnataka, where ARTPARK collaborated with the Karnataka State Disaster Management Authority (KSNDMC) and the state health department. Later, the



Dr Raunaq Pradhan, product manager, ARTPARK, Indian Institute of Science (IISc), Bengaluru

collaborations extended to include the India Meteorological Department (IMD), ICMR and the National Vector Borne Disease Control Programme (NVBDCP) for expansion at a national scale.

Karnataka offered strong historical weather and disease datasets, allowing researchers to build and validate forecasting models while working closely with government agencies responsible for outbreak management.

The first layer in the ForecastHub is a data factory that continuously ingests climate information from sources such as ERA5 and IMD, along with historical disease data aligned with government of India's Integrated Health Information Platform (IHIP).

Once collected, all datasets are standardised, validated and stored in version-controlled repositories to ensure consistent model training.

The second layer hosts forecasting models. Initial deployments rely on negative binomial regression and time-series forecasting models, while newer AI models, including Google's foundational models, are being evaluated. The architecture allows additional forecasting models to be integrated without redesigning the platform.

Turning predictions into action

Forecast outputs are translated into risk classifications instead of raw numbers.